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ADI1221– PRINCIPLES OF ARTIFICIAL INTELLIGENCE

Loan/Credit Eligibility Reasoning Engine Using FQL-Based Rule Inference on Live Application Data

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PRESENTATION OVERVIEW



1. Problem Statement
2. Objectives
3. Literature Survey
4. AI technique selection with justification
5. System Architecture
6. Dataset and API identification
7. Project Planning
8. Closing Statement



PROBLEM STATEMENT



Background

- Traditional loan approval systems rely on manual verification or fixed rule-based methods, making decisions slow, inconsistent, and prone to human errors.

Current Challenges

- High processing time
- Manual document verification
- Inconsistent decision-making
- Fraudulent applications
- Limited explainability
- Increasing loan defaults
- Need
- Develop an AI-powered reasoning engine that evaluates live applicant data using FQL rules.



OBJECTIVES



- To study the existing FQL-based loan eligibility concepts as a reference.
- To understand how Fuzzy Logic and IF–THEN rules are used in credit assessment.
- To refer existing methods for analyzing income, credit score, debt, and loan amount.
- To use Low, Medium, and High fuzzy conditions for handling uncertain data.
- To develop a loan eligibility reasoning system based on the studied reference concepts.
- To integrate live application data through APIs for real-time evaluation.
- To improve the existing approach by providing flexible and explainable eligibility decisions.



LITERATURE SURVEY



Existing Research

- Credit Scoring Models – Use income, credit history, debt, and credit score to evaluate loan risk.
- Fuzzy Logic – Handles uncertain financial values such as Low Income, Moderate Debt, and Good Credit Score.
- Fuzzy Rule-Based Systems – Use IF–THEN rules to combine different applicant factors and make eligibility decisions.
- FQL-Based Reasoning – Processes fuzzy data using linguistic values such as Low, Medium, and High.



References

- 1. K. C. Shekhar & Lekshmy Shekhar, Banking Theory and Practice, Vikas Publishing House.
- 2. C. K. Shukla, Banking and Financial Services, Himalaya Publishing House.
- 3. Research papers on Fuzzy Logic and Machine Learning for Credit Risk/Loan Approval – IEEE Xplore.
- 4. Reserve Bank of India (RBI) – Banking and lending guidelines.
- 5. Kaggle – Loan approval/eligibility datasets.



AI TECHNIQUE SELECTION & JUSTIFICATION



FQL is used to query and reason over fuzzy applicant data using rules, allowing the system to make flexible loan eligibility decisions instead of using only strict Yes/No conditions.

Example :

FQL in Loan Eligibility

Applicant Details:

Income = ₹40,000 → Medium/High

Credit Score = 720 → Good

Debt = ₹8,000 → Low



Why FQL is Used?

- Handles uncertain data in loan applications.
- Uses Low, Medium, High type fuzzy values.
- Evaluates income, credit score, debt, etc.
- Supports rule-based decision making.
- Gives flexible eligibility decisions instead of only Yes/No.
- Helps improve the reasoning and accuracy of the loan eligibility system.



Rule 1

- IF
- Credit Score > 750
- AND Income $> ₹50,000$
- AND Existing Loans = 0
- THEN
- Eligible

Rule 2

- IF
- Credit Score < 600
- THEN
- Reject



Rule 3

- IF
- Debt-to-Income Ratio $> 45\%$
- THEN
- High Risk



Customer Financial Stability Assessment



- Analyses the customer's financial condition over time.
- Considers income, debt, repayment behaviour, and credit score trends.
- Classifies the customer as Stable, Moderate, or Unstable.
- Helps predict the customer's future financial risk, not just their current eligibility.

Key Factors

- Income Trend – Increasing / Stable / Decreasing
- Debt Trend – Decreasing / Stable / Increasing
- Repayment Behaviour – On-time / Delayed
- Credit Score Trend – Improving / Stable / Declining

SYSTEM ARCHITECTURE





DATASET & API IDENTIFICATION

Required Features



Dataset:

- Loan/Credit application dataset is used.
- Contains applicant details such as income, age, credit score, loan amount, employment status, debt, and repayment history.
- Used to create and test FQL-based eligibility rules.
- Dataset can be obtained from Kaggle or other public financial datasets.

API:

- A Loan Application API provides live applicant data.
- API sends income, credit score, loan amount, and debt to the reasoning engine.
- FQL rules process the received data.
- The system returns the final loan eligibility status and reasoning.



PROJECT PLANNING



Phase 1

- Requirement Analysis
- Study banking rules
- Collect requirements

Phase 2

- Dataset Collection
- Loan records
- Credit information
- Customer profiles

Phase 3

- Data Preprocessing
- Missing value handling
- Data normalization
- Feature engineering

Phase 4

- FQL Rule Development
- Banking rules
- Credit rules
- Risk rules



Phase 5

- Inference Engine Development
- Rule parser
- Decision engine
- Risk analyzer

Phase 6

- API Integration
- Credit score API
- Banking API
- Identity verification



Phase 7

- Testing
- Accuracy
- Performance
- Security
- Explainability

Phase 8

- Deployment
- Cloud deployment
- Dashboard
- Monitoring



EXPECTED OUTCOME



- Develop an FQL-based Loan/Credit Eligibility Reasoning Engine.
- Analyze applicant details such as income, credit score, debt, and loan amount.
- Apply fuzzy rules to handle uncertain or borderline values.
- Generate a Loan Eligibility Score.
- Classify applicants as High, Medium, or Low eligibility.
- Provide a clear reasoning/explanation for the eligibility decision.
- Support faster and more consistent loan decision-making.



CLOSING STATEMENT



The proposed Loan/Credit Eligibility Reasoning Engine combines Formal Query Language (FQL)-based rule inference with live financial data to deliver accurate, transparent, and explainable loan decisions. By integrating AI reasoning, real-time APIs, and risk assessment, the system improves efficiency, reduces fraud, ensures regulatory compliance, and supports faster, fairer credit approvals. This approach demonstrates how Explainable AI can modernize financial decision-making while maintaining trust and accountability.



Thank You